Numerical Methods for Convection-Dominated Problems

Central & Upwind Finite Difference Schemes

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Outline

Introduction

Convection–diffusion equations appear across science and engineering:

- **Fluid mechanics** — heat and mass transport in flowing fluids
- **Environmental science** — contaminant spread in groundwater
- **Atmospheric modelling** — pollutant dispersion in air
- **Finance** — Black–Scholes option pricing equations

Model Problem

$$-\varepsilon\, u''(x) + b\, u'(x) = f(x), \quad x \in (0,1), \qquad u(0) = \alpha,\ u(1) = \beta.$$

The Core Challenge

When $0 < \varepsilon \ll |b|$, the solution develops a **thin boundary layer** near the outflow boundary. Standard numerical methods on coarse meshes produce **spurious oscillations**.

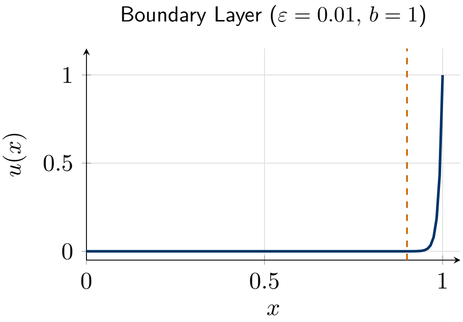
Why This Problem Is Hard

Competing physical effects:

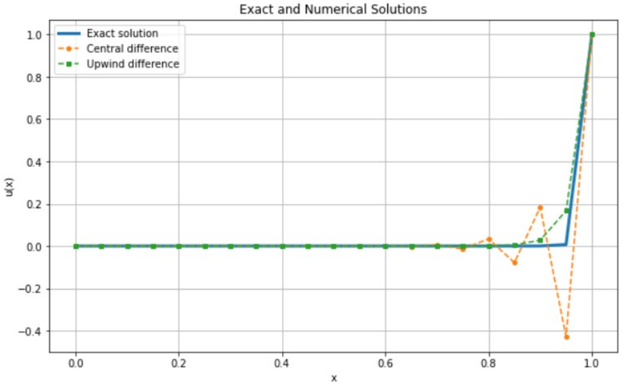
- **Diffusion** $-\varepsilon u''$: smooths gradients, spreads information in *all* directions.
- **Convection** $b\,u'$: transports information *upstream* only.

Consequence:

Setting $\varepsilon = 0$ gives a first-order ODE requiring only *one* BC. The mismatch forces a sharp layer of width $O(\varepsilon/|b|)$.



Numerical Evidence: Coarse Mesh Failure



Coarse grid behaviour:

- **Exact solution** rises smoothly in the layer.
- **Central** produces large **oscillations** near $x = 1$, dropping to -0.4.
- **Upwind** stays non-negative but visibly smears the layer.

Key Takeaway

When $\mathrm{Pe}_h > 1$, central differencing is **physically unreliable**.

Objectives

Mathematical Goals

O1. Derive the **exact solution** and quantify the boundary layer width.

O2. Establish the **weak formulation** in $H_0^1(0,1)$ and prove uniqueness via **Lax–Milgram**.

O3. Derive **energy estimates** and verify the **maximum principle**.

Numerical Goals

O4. Construct the **central** and **upwind** difference schemes on a uniform grid.

O5. Perform **truncation error analysis** and interpret numerical diffusion.

O6. Validate convergence rates $O(h^2)$ and $O(h)$ via a **mesh refinement study**.

The Strong Form of the Problem

1D Stationary Convection–Diffusion BVP

$$-\varepsilon u''(x) + b\,u'(x) = f(x), \qquad x \in (0,1), \qquad u(0) = \alpha, \quad u(1) = \beta.$$

Parameters:

- $\varepsilon > 0$: diffusion coefficient
- $b \in \mathbb{R}$: convection velocity
- $f(x)$: source term

Competing effects:

- $-\varepsilon u''$: **diffusion**
- $b\,u'$: **convection**

Convection-Dominated Regime

When $0 < \varepsilon \ll |b|$, a **boundary layer** of width $O(\varepsilon/|b|)$ forms near the outflow boundary.

Exact Solution of the Model Problem

Homogeneous Problem

$-\varepsilon u'' + b\, u' = 0$ on $(0,1)$, $\quad u(0)=0,\ u(1)=1.$

Solution steps:

1. Let $v = u'$. Then $\varepsilon v' - bv = 0 \ \Rightarrow\ v(x) = C_1 e^{bx/\varepsilon}$.
2. Integrate: $u(x) = Ae^{bx/\varepsilon} + B$.
3. BCs: $u(0)=0 \Rightarrow B = -A;\quad u(1)=1 \Rightarrow A = \left(e^{b/\varepsilon}-1\right)^{-1}$.

Closed-Form Exact Solution

$$u(x) = \frac{e^{bx/\varepsilon}-1}{e^{b/\varepsilon}-1}.$$

Observation: For small ε, $e^{b/\varepsilon} \gg 1$ — solution stays near zero on $[0,1)$ and rises sharply to 1 near $x=1$.

Weak Formulation

Homogeneous BCs: $u(0) = u(1) = 0$; test functions $v \in H_0^1(0,1)$.

Multiply by v, integrate, and integrate by parts:

$$\varepsilon \int_0^1 u'v'\,dx + b\int_0^1 u'v\,dx = \int_0^1 fv\,dx.$$

Weak Problem

Find $u \in H_0^1(0,1)$ such that

$$\underbrace{\varepsilon \int_0^1 u'v'\,dx + b\int_0^1 u'v\,dx}_{a(u,v)} = \underbrace{\int_0^1 fv\,dx}_{L(v)} \qquad \forall\, v \in H_0^1(0,1).$$

Well-Posedness via Lax–Milgram

Boundedness of $a(\cdot,\cdot)$

$$|a(u,v)| \leq (\varepsilon + |b|)\|u\|_{H^1}\|v\|_{H^1}.$$

Boundedness of $L(\cdot)$

$$|L(v)| \leq \|f\|_{L^2}\|v\|_{H^1}.$$

Coercivity of $a(\cdot,\cdot)$

Since $v \in H_0^1$: $\int_0^1 v'v\,dx = 0$.

By Poincaré:

$$a(v,v) = \varepsilon\|v'\|_{L^2}^2 \geq \frac{\varepsilon}{C_P^2+1}\|v\|_{H^1}^2.$$

Lax–Milgram Theorem

$a(\cdot,\cdot)$ is **bounded** and **coercive**, $L(\cdot)$ is **bounded** $\Rightarrow$ **unique** weak solution $u \in H_0^1(0,1)$.

Energy Estimate & Maximum Principle

Energy Estimate

Setting $v = u$:

$$\|u'\|_{L^2} \leq \frac{C_P}{\varepsilon}\|f\|_{L^2}, \qquad \|u\|_{L^2} \leq \frac{C_P^2}{\varepsilon}\|f\|_{L^2}.$$

Bound grows like $1/\varepsilon$ — gradient blows up as $\varepsilon \to 0$.

Maximum Principle

If $f \leq 0$ and u attains an interior max at x_0:

$u'(x_0) = 0$,
$u''(x_0) \leq 0 \;\Rightarrow\; f(x_0) = -\varepsilon u''(x_0) \geq 0$.

Contradiction $\Rightarrow$ max controlled by BCs.

Why Numerics Are Hard

Layer width $\sim \varepsilon$ requires $h \ll \varepsilon$. Standard schemes on coarse meshes fail $\Rightarrow$ **spurious oscillations**.

Uniform Grid and Derivative Approximations

Uniform Grid

$$h = \frac{1}{N}, \quad x_i = ih,$$
$$i = 0, 1, \ldots, N.$$

Unknowns: $U_1, \ldots, U_{N-1}$.

BCs: $U_0 = \alpha$, $U_N = \beta$.

Taylor-Derived Approximations

2nd derivative (central, $O(h^2)$):

$$u''(x_i) \approx \frac{U_{i+1} - 2U_i + U_{i-1}}{h^2}.$$

1st derivative — central ($O(h^2)$):

$$u'(x_i) \approx \frac{U_{i+1} - U_{i-1}}{2h}.$$

1st derivative — upwind ($O(h)$, $b > 0$):

$$u'(x_i) \approx \frac{U_i - U_{i-1}}{h}.$$

Central Difference Scheme

$$-\varepsilon \frac{U_{i+1} - 2U_i + U_{i-1}}{h^2} + b \frac{U_{i+1} - U_{i-1}}{2h} = f_i.$$

Central Difference Stencil

$$a_C U_{i-1} + b_C U_i + c_C U_{i+1} = f_i,$$

$$a_C = -\frac{\varepsilon}{h^2} - \frac{b}{2h}, \qquad b_C = \frac{2\varepsilon}{h^2}, \qquad c_C = -\frac{\varepsilon}{h^2} + \frac{b}{2h}.$$

Stability Issue: Mesh Péclet Number

$$\mathrm{Pe}_h = \frac{|b|h}{2\varepsilon}.$$

When $\mathrm{Pe}_h > 1$: c_C changes sign $\Rightarrow$ loss of diagonal dominance $\Rightarrow$ **spurious oscillations**.

Upwind Difference Scheme

For $b > 0$, use the *backward* (upwind) difference for u':

$$-\varepsilon \frac{U_{i+1} - 2U_i + U_{i-1}}{h^2} + b \frac{U_i - U_{i-1}}{h} = f_i.$$

Upwind Difference Stencil

$$a_U\, U_{i-1} + b_U\, U_i + c_U\, U_{i+1} = f_i,$$

$$a_U = -\frac{\varepsilon}{h^2} - \frac{b}{h}, \qquad b_U = \frac{2\varepsilon}{h^2} + \frac{b}{h}, \qquad c_U = -\frac{\varepsilon}{h^2}.$$

Why It Works

$a_U, c_U \leq 0$ and $b_U \geq |a_U| + |c_U|$ for all $h > 0$ $\Rightarrow$ strict diagonal dominance $\Rightarrow$ **unconditionally stable** and **monotone**.

Matrix Form & Boundary Conditions

Incorporating BCs into RHS:

- At $i = 1$: subtract $a \,.\, \alpha$ from f_1.
- At $i = N - 1$: subtract $c \,.\, \beta$ from f_{N-1}.

Tridiagonal Structure

Both schemes produce **tridiagonal** matrices solved by the Thomas algorithm in $O(N)$ operations.

Linear system: $\mathbf{AU} = \mathbf{F}$

$$A = \begin{pmatrix} b. & c. & & \\ a. & b. & c. & \\ & \ddots & \ddots & \ddots \\ & & a. & b. \end{pmatrix}$$

Local Truncation Error

Central Scheme — $O(h^2)$

Both approximations are $O(h^2)$:

$$\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} = u'' + O(h^2),$$

$$\frac{u_{i+1} - u_{i-1}}{2h} = u' + O(h^2).$$

$$\boxed{\tau_i^{(C)} = O(h^2)}$$

Upwind Scheme — $O(h)$

Backward difference:

$$\frac{u_i - u_{i-1}}{h} = u' - \frac{h}{2}u'' + O(h^2).$$

$$\boxed{\tau_i^{(U)} = O(h)}$$

Effective diffusion:

$$\varepsilon_{\text{eff}} = \varepsilon + \tfrac{bh}{2}.$$

Accuracy–Stability Trade-Off

Upwinding stabilises via **numerical diffusion** $\frac{bh}{2}$ — smears the layer but eliminates oscillations.

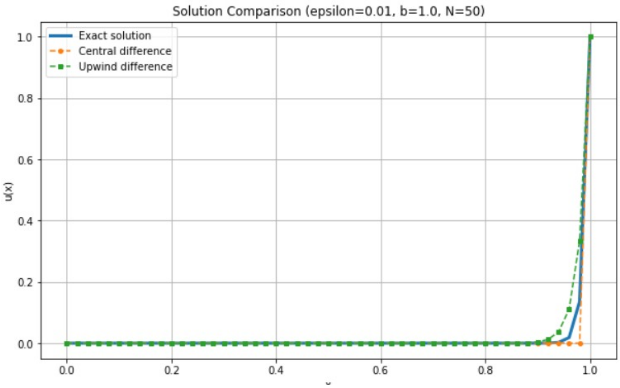
Scheme Comparison

Property	Central Difference	Upwind Difference
Order of accuracy	$O(h^2)$	$O(h)$
Truncation error type	Dispersion	Numerical diffusion
Stable when	$\mathrm{Pe}_h \leq 1$	Always
Spurious oscillations	Yes (large Pe_h)	No
Monotone solution	Not guaranteed	Guaranteed
Boundary layer capture	Poor (coarse mesh)	Smeared but stable

Practical Rule

Central: preferred when $h \ll 2\varepsilon/|b|$ (fine mesh). **Upwind**: preferred when stability is paramount.

Numerical Results: Fine Mesh ($N = 50$, $\varepsilon = 0.01$, $b = 1$)



With $N = 50$ grid points:

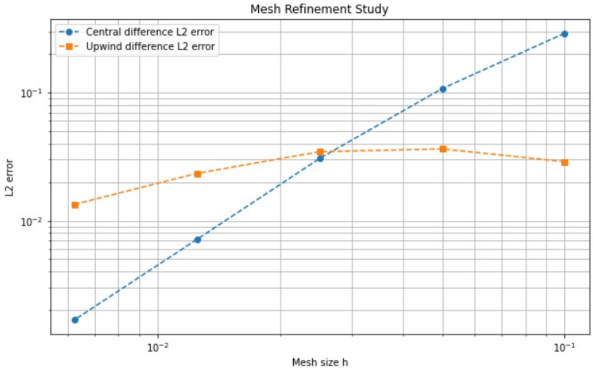
$$h = 0.02, \quad \mathrm{Pe}_h = \frac{0.02}{2 \times 0.01} = 1.$$

- **Exact solution** captures the sharp boundary layer.
- **Central** now stable at $\mathrm{Pe}_h = 1$, closely tracks exact.
- **Upwind** still visibly smeared.

Observation

At $\mathrm{Pe}_h = 1$, central differencing recovers good behaviour.

Numerical Results: Mesh Refinement Study



L^2 error vs. mesh size h:

- **Central**: error decreases steeply, consistent with $O(h^2)$.
- **Upwind**: plateaus on coarse meshes, then decays at $O(h)$.

Confirmed Convergence Rates

Scheme	Rate p
Central	≈ 2
Upwind	≈ 1

Summary

Theory established:

- Strong form & boundary layer phenomenon
- Exact solution via ODE analysis
- Weak formulation in $H_0^1(0,1)$
- Boundedness & coercivity of $a(\cdot,\cdot)$
- Existence & uniqueness via Lax–Milgram
- Energy estimate & maximum principle

Numerics established:

- Uniform finite difference grid
- Central scheme ($O(h^2)$, conditionally stable)
- Upwind scheme ($O(h)$, unconditionally stable)
- Tridiagonal matrix assembly
- Truncation error & numerical diffusion
- Convergence rates confirmed numerically

Overarching Message

The convection-diffusion problem is well-posed but **numerically challenging** when $\varepsilon \ll |b|$. Upwinding provides stability; higher-order methods (SUPG, exponential fitting) recover accuracy.

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Thank You

Questions & Discussion

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